

Spectral methods for singular boundaries: a survey

Comparing GYRE (stellar pulsation) and Dedalus v3 (general spectral) against the Liouville SL-anelastic design

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1. Background and problem statement

1.1 Motivation

This project's `pseudo_spectral` solver currently handles uniform-density 2D incompressible Navier–Stokes on a doubly periodic domain. Extending it to **anelastic stellar convection** promotes the pressure equation from a constant-coefficient Laplace operator to a variable-coefficient elliptic equation:

$$\nabla \cdot \left(\frac{1}{\rho_0(y)} \nabla p \right) = f(x, y)$$

The original design (`docs/anelastic_SL_spectral_design.md`, 2026-05-01) proposes a Liouville normal-form substitution $\hat{p} = \sqrt{\rho_0} q$ to reduce the variable-coefficient equation to a Schrodinger-type form:

$$q'' + W(y) q - k_x^2 q = g, \quad W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}$$

Solving the eigenproblem for $[d^2/dy^2 + W]$ yields a Sturm–Liouville (SL) basis (μ_n, ψ_n) ; all k_x modes share the same basis, and Poisson inversion reduces to a per-mode scalar division.

1.2 The unresolved problem: the $\rho \rightarrow 0$ singularity

Phase 0 validation (`docs/anelastic_sl_phase0_2026-05-02.md`) shows:

- The Lane–Emden $n = 1.5$ surface $\rho \rightarrow 0$ drives $|W| \rightarrow 1.35 \times 10^6$ (divergent).
- Truncation to $\rho > 0.01$ (i.e. $r < 0.94$) makes the eigenproblem solvable.
- SL–Poisson end-to-end on a manufactured solution: $\text{err}_{L^2} = 3.7 \times 10^{-6}$.
- **Convergence order** ~ -2.4 (**algebraic**), not the expected exponential.
- Phase 0 E3 confirms: a smooth Gaussian $\rho(y)$ restores exponential convergence. The algebraic rate is therefore entirely driven by the singular boundary.

Core observation. The Liouville substitution does not eliminate the $\rho \rightarrow 0$ singularity; it merely transports it from a variable-coefficient elliptic equation to a Schrodinger potential. We still need to address it.

1.3 Purpose of this survey

The quantum-mechanics community (Coulomb potentials) and the stellar-pulsation community (adiabatic oscillations) have accumulated 60 years of standard techniques for exactly this class of singularity. This report surveys two representative open-source codes:

1. **GYRE** (Townsend 2013; Fortran)

- Astrophysical stellar pulsation (adiabatic oscillations).
- Structurally isomorphic to our problem: a radial SL eigenproblem inside a star.
- Repo: <https://github.com/rhdtownsend/gyre>

2. **Dedalus v3** (Burns et al. 2020; Python)

- General-purpose spectral PDE framework.
- Covers Chebyshev / Jacobi / Ball / Disk bases.
- Repo: <https://github.com/DedalusProject/dedalus>

The central question: **how do these two communities treat $\rho \rightarrow 0$ -type boundary singularities, and what can this project borrow?**

2. GYRE's strategy

2.1 Overall philosophy: pre-designed variable transformations

GYRE does not use a Liouville $\sqrt{\rho}$ substitution. It employs a **non-dimensional geometric substitution**:

$$y_1 = x^{2-\ell} \cdot \frac{\xi_r}{r}, \quad y_2 = x^{2-\ell} \cdot \frac{P'}{\rho g r}$$

where ξ_r is the radial displacement and P' is the pressure perturbation.

Key point. This substitution absorbs geometric singularities (origin $r = 0$ and surface $\rho \rightarrow 0$) **into the definition of the evolved variables**, so that y_1, y_2 remain bounded throughout the radial domain.

Reference: docs/source/ref-guide/osc-equations/dimless-form.rst:24-32

2.2 Surface boundary: analytic asymptotic form

GYRE does not impose Dirichlet or Neumann conditions at the surface. It uses **physically motivated asymptotic boundary conditions**:

- **DZIEM mode** (Dziembowski 1971):

$$\left\{ 1 + \frac{1}{V}[\dots] \right\} y_1 - y_2 + \dots = 0$$

- **UNNO mode** (Unno et al. 1989): Eddington approximation.

- **VACUUM mode:** minimal form, $y_2 = 0$.
- **JCD mode:** Jorgensen-Christensen-Dalsgaard detailed form.

References: src/eqns/rad/rad_obound_m.fypp:102-133, src/eqns/rad/gyre/OB_dziem.inc:1-3

These conditions **all assume a physical limit exists at the surface**, rather than a divergence. This is the deep reason GYRE can sidestep the $\rho \rightarrow 0$ singularity: its non-dimensional variables y_1, y_2 are finite at the surface regardless of what W is doing.

2.3 Grid refinement: ODE-eigenvalue criteria

src/grid/grid_refine_m.fypp implements adaptive log-grid bisection. The refinement criterion is driven by ODE eigenvalues χ :

```
! Mechanical / thermal / structural criteria combined
split(j) = split_mech(...) .OR. split_therm(...) .OR. split_struct(...)
```

In $\ln(x)$ space, $\text{dln}x * |\chi_{\text{imag}}|$ (wavelength criterion) and $|\chi_{\text{real}}|^{-1}$ (decay length criterion) are evaluated, so **grid points are automatically added where oscillations or decay are rapid**.

References: src/grid/grid_refine_m.fypp:78-80, 214-226

2.4 Polytropic model treatment

The Lane–Emden solver uses a **fourth-order Taylor series + LSODAR event detection**:

```
X_BEG = sqrt(EPSILON(0._RD))    ! ~ 1e-8
y_exp(1) = 1 - x_exp**2/6 + n_poly*x_exp**4/120    ! theta(z) Taylor expansion
```

The key trick: the Lane–Emden RHS has a $1/x^2$ factor that looks singular at $x = 0$, but using $\theta \propto 1 - z^2/6$ near zero makes it finite.

Reference: src/poly/lane_emden_m.fypp:59, 109-110

2.5 Discretization: Magnus exponential integrator

No Chebyshev or FEM. GYRE uses **Gauss–Legendre Magnus integrators** (GL2 / GL4 / GL6):

```
! Diagonalize the ODE as dy_i/dx = lambda_i * y_i, then integrate exponentially
```

Reference: src/diff/magnus_gl2_diff_m.fypp:20-56

2.6 Lessons from GYRE

Technique	Portability to this project
Pre-designed non-dimensional variables that absorb singularity	★ ★ ★ Most central; worth adopting
Analytic asymptotic BC (Dziembowski)	★★ Useful reference; requires physical insight
ODE-eigenvalue adaptive grid	★ Complex; not urgent
Magnus exponential integrator	– Orthogonal to spectral methods

3. Dedalus v3's strategy

3.1 Overall philosophy: weighted Jacobi bases

All of Dedalus's radial bases belong to the Jacobi family $J_n^{(a,b)}(x)$. The basis functions carry an intrinsic weight $(1-x)^a(1+x)^b$.

Key insight. Endpoint singularities can be **fully encoded in the weight exponents** a, b .

- ChebyshevT: $a = b = -1/2$, absorbs $(1 \pm x)^{-1/2}$ singularities.
- Ultraspherical(alpha): $a = b = \alpha - 1/2$, tunable.
- Jacobi(a,b): arbitrary (a, b) .

Reference: dedalus/core/basis.py:435–515

3.2 Ball/Disk basis: regularity at $r = 0$

BallBasis and DiskBasis use a **k parameter** to encode an r^k embedding:

```
# BallRadialBasis.__init__(..., k=2):
#   basis functions automatically carry an  $r^k$  factor
#   Zernike polynomials *  $r^{(\alpha + k + \ell + \text{regtotal})}$ 
```

regularity_allowed() automatically filters out physically disallowed $(\ell, m, \text{regularity})$ combinations.

Reference: dedalus/core/basis.py:3917–4087, 3531–3594

3.3 Boundary conditions: tau / lift method

Instead of Dirichlet, Dedalus uses a lift method:

```
problem.add_equation("lap(f) + lift(tau) = -f**n")
problem.add_equation("f(r=1) = 0")    # the tau term absorbs boundary residual
```

LiftJacobi.build_polynomial() picks a high-order polynomial ϕ_N ; the boundary residual is projected onto it, avoiding pollution of the eigen-space.

References: dedalus/core/basis.py:790–815, examples/nlbvp_ball_lane_emden/lane_emden.py:67–71

3.4 Variable-coefficient operator matrices

`operator_matrix()` pre-computes variable-coefficient operators on Zernike / Jacobi bases, including r^{-k} singular operators, without manual regularization:

Reference: `dedalus/core/basis.py:4044-4056`

3.5 Things Dedalus does not do

- No Kosloff–Tal-Ezer (coordinate stretching) mapping.
- No Liouville $\sqrt{\rho}$ substitution.
- No explicit cutoff.

Everything relies on the Jacobi weight absorbing singular behavior.

3.6 Lessons from Dedalus

Technique	Portability to this project
Jacobi basis with chosen (a, b) weights	★ ★ ★ Core idea; structurally similar to GYRE’s substitution
r^k embedding for Ball/Disk	★★ Not immediately relevant (we’re 2D plane)
Tau / lift method for BC	★★ Cleaner than FD Dirichlet
Variable-coefficient operator caching	★ Engineering; later phase

4. Comparison: SL vs GYRE vs Dedalus

4.1 Three different philosophies for “variable coefficient + singular boundary”

Aspect	Project SL (current)	GYRE	Dedalus
Discretization	Interior FD	Magnus exponential	Jacobi spectral
Variable substitution	$\hat{p} = \sqrt{\rho} q$	$y_1 = x^{2-\ell} \xi / r$	None (basis carries weight)
Singularity absorption	None; truncate $\rho > 0.01$	Variable definition absorbs	$(1 - x)^a (1 + x)^b$ absorbs
Boundary condition	Dirichlet (interior FD)	Analytic asymptotic (Dziembowski)	Tau / lift
Surface singularity	Regexp to truncation	Variables finite at surface	Basis handles intrinsically
Convergence order	Algebraic ~ -2.4	Analytic (Magnus high-order)	Exponential

4.2 Why is the Liouville $\sqrt{\rho}$ substitution not enough?

The Liouville substitution converts

$$\frac{d}{dy} \left[\frac{1}{\rho} \frac{dp}{dy} \right] = f$$

into

$$q'' + Wq = g,$$

but $W = \rho''/(2\rho) - 3(\rho')^2/(4\rho^2)$ **still diverges as $\rho \rightarrow 0$** . We have only moved the singularity; we have not destroyed it.

GYRE and Dedalus instead **ensure no singular term ever appears** in the discretized operator:

- GYRE: the definitions $y_1 = x^{2-\ell}\xi/r$ and $y_2 = x^{2-\ell}P'/(\rho gr)$ force the entire LHS to be finite at the surface.
- Dedalus: Jacobi basis $J_n^{(\alpha,\beta)}(x)$ carries $(1-x)^\alpha$ factors, and if α matches the decay exponent of ρ , the variable-coefficient ODE is analytically solvable in the weighted inner product.

5. Three possible paths forward for this project

5.1 Path A: keep Liouville SL, add a GYRE-style substitution

Motivation. Preserve the SL selling point: one set of (μ_n, ψ_n) serving both Poisson inversion and the g-mode spectrum.

How. In addition to $\hat{p} = \sqrt{\rho} q$, add a second substitution $\tilde{\phi} = r^\beta \phi$, choosing β so that the induced operator $\tilde{W}$ remains finite as $\rho \rightarrow 0$.

Pros. Physical elegance preserved; the g-mode spectrum still emerges automatically as an eigenspectrum of the basis.

Cons. Derivation-heavy; requires careful verification that $\tilde{W}$ is smooth.

5.2 Path B: switch to Jacobi–Galerkin

Motivation. The Dedalus approach is numerically validated by our own Phase 0 E3 (smooth Gaussian ρ restores exponential convergence).

How. Drop the Liouville substitution and work directly in a Jacobi basis $J_n^{(\alpha,\beta)}(y)$. Pick $\alpha = 1.5$ to match the Lane–Emden $n = 1.5$ surface decay exponent.

Pros. Simplest engineering; Dedalus can be imported as a library for validation.

Cons. The g-mode spectrum now requires a **separate** generalized eigenproblem $L\phi = \omega^2 M\phi$. It is no longer a by-product of the basis. The “unified basis” pitch of the original design weakens.

5.3 Path C: do both A and B, write a method-comparison paper

Motivation. The numerical comparison is itself a contribution.

How.

1. Phase 1 in parallel: Boussinesq baseline using Liouville SL and Jacobi.
2. Compare err_{L^2} , wall-clock, and g-mode accuracy.
3. Publish a pure method-comparison paper (JCP fits well).

Pros. Most defensible; results cannot be wrong.

Cons. ~ 1.5 – 2 x the workload; less focused narrative.

6. Recommendations

6.1 Path selection is a function of the paper’s angle

Paper angle	Recommended path
“GPU-efficient variable-coefficient Poisson” (JCP methods)	Path B (Jacobi)
“Unified spectral basis for Poisson + g-mode spectroscopy” (ApJS astro)	Path A (SL + substitution)
“Method comparison for stratified flows” (JCP comparison)	Path C

6.2 Technical risk assessment

Path A risk. The substitution $\phi \rightarrow r^\beta \phi$ requires β to match the Lane–Emden surface decay exponent of 1.5 with high accuracy. If mismatched, the singularity residue remains and the convergence is still algebraic. **Recommended: run a Python verification first to confirm that $\beta = 1.5$ restores exponential convergence.**

Path B risk. Dedalus’s EVP solver may need careful tau-parameter tuning on stiff variable-coefficient problems. That said, Dedalus ships a Lane–Emden Ball example (`examples/nlbvp_ball_lane_emden/`), indicating the framework already handles analogous singularities. Engineering risk is low.

Path C risk. Largest workload, but risk is distributed.

6.3 Suggested execution order

Phase 0 ext+ (1–2 days, pure Python)

1. **Use Dedalus as a library, solve Lane–Emden Poisson** (quick validation of Path B).
 - Expected: $\text{err}_{L^2} \leq 10^{-10}$, exponential convergence.

- If confirmed: Path B has zero technical risk.
2. **Keep Liouville SL, add $\phi \rightarrow r^\beta \phi$ substitution** (Path A exploration).
 - Verify that matching β restores exponential convergence.
 - If confirmed: Path A has zero technical risk.
 3. **Both results on the table, then decide Phase 1’s main path.**

This means Phase 1’s code commitment will be backed by numerical evidence, not by a hunch.

7. Conclusions

7.1 Main findings

1. **Neither GYRE nor Dedalus uses a Liouville $\sqrt{\rho}$ substitution.** They absorb the singularity at the variable-definition or basis-weight level, from the start, rather than regularizing it after the fact.
2. **The Phase 0 algebraic —2.4 convergence is fixable.** E3 already demonstrated exponential convergence for a smooth Gaussian ρ . The problem is that our basis / variable substitution is not aligned with the singularity, not that the SL method itself is fundamentally limited.
3. **The g-mode selling point behaves very differently in each path.**
 - Liouville SL: g-mode spectrum is a free by-product (elegant).
 - Jacobi–Galerkin: g-mode requires a separate EVP (standard but loses the “unified basis” pitch).

7.2 Corrections to the original SL–anelastic design

The mathematics in docs/anelastic_SL_spectral_design.md is **entirely correct** (the Liouville derivation in sect. 3–4 and the SL diagonalization in sect. 4.2 are both rigorous). The engineering assumptions, however, are **overly optimistic**:

- Section 8.2’s “smooth $W \rightarrow$ exponential, singular $W \rightarrow$ algebraic” dichotomy is now empirically confirmed by Phase 0.
- Section 8.1’s surface-singularity TODO has been Phase 0-confirmed as the primary convergence obstacle.
- Section 7’s “three-community gap” narrative remains correct, but the reason needs revision: it is not “no one thought of SL”; it is **“other communities have already adopted simpler approaches that sidestep the Liouville substitution”**.

Proposed revisions:

- Upgrade sections 8.1 and 8.2 from TODOs to **primary challenges**.
- Split Phase 1 into B-quick-validation and A-deep-exploration in parallel.
- Adjust the paper angle from “SL is the optimal method” to “SL vs Jacobi: a method comparison”.

7.3 Direction forward

Short term (Phase 0 ext+, 1–2 days): execute section 6.3 – Dedalus Lane–Emden validation plus Liouville + r^β substitution experiments.

Medium term (Phase 1 decision point): based on Phase 0 ext+ results, choose A / B / C.

Long term (Phase 2-3): Boussinesq to anelastic physics extension, independent of basis choice. g-mode spectrum cross-validation against GYRE (using Lane–Emden $n = 1.5$ as a common benchmark) will provide the final physical fidelity check.

Appendix A: Key GYRE files

- `src/eqns/rad/rad_obound_m.fypp` – outer boundary condition dispatch
- `src/eqns/rad/rad_vars_m.fypp` – non-dimensional variables y_1, y_2
- `src/grid/grid_refine_m.fypp` – adaptive grid refinement
- `src/poly/lane_emden_m.fypp` – Lane–Emden integrator
- `src/diff/magnus_gl2_diff_m.fypp` – Magnus GL2/GL4/GL6
- `docs/source/ref-guide/osc-equations/dimless-form.rst` – theoretical formulation

Appendix B: Key Dedalus v3 files

- `dedalus/core/basis.py:435–515` – Jacobi / ChebyshevT / Ultraspherical
- `dedalus/core/basis.py:3917–4087` – BallBasis / DiskBasis
- `dedalus/core/basis.py:790–815` – Lift / Tau method
- `dedalus/core/basis.py:4044–4056` – variable-coefficient operator matrices
- `examples/nlbvp_ball_lane_emden/lane_emden.py` – Lane–Emden example
- `examples/evp_disk_pipe_flow/pipe_flow.py` – Disk EVP example

Appendix C: References

- **Dziembowski (1971)**: Non-radial oscillations of evolved stars. I. *Acta Astronomica* 21, 289.
- **Unno et al. (1989)**: *Nonradial Oscillations of Stars*, 2nd ed., University of Tokyo Press.
- **Tassoul (1980)**: Asymptotic approximations for stellar nonradial pulsations. *ApJS* 43, 469.
- **Townsend & Teitler (2013)**: GYRE: an open-source stellar oscillation code. *MNRAS* 435, 3406.
- **Burns et al. (2020)**: Dedalus: a flexible framework for numerical simulations with spectral methods. *PhRvR* 2, 023068.
- **Kosloff & Tal-Ezer (1993)**: A modified Chebyshev pseudospectral method. *JCP* 104, 457.
- **Boyd (2001)**: *Chebyshev and Fourier Spectral Methods*. Dover Publications.